

Breathing Techniques and Breath-Regulation Strategies in Sport and Performance Contexts: A Scoping Review

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Abstract

Background: Breathing techniques are widely used in sport and performance settings, but the evidence spans many techniques, populations, outcomes, and reporting styles. **Objective:** This scoping review maps evidence on non-device breathing and breath-regulation strategies, integrating bibliometric, participant, intervention, efficacy, feasibility, and safety data. **Methods:** The review followed PRISMA-ScR principles. Sixty-five reports were charted and synthesized descriptively; exploratory quantitative syntheses were conducted when outcome data were extractable. Efficacy effects were standardized so that positive Hedges' g values favored the breathing intervention. Safety and dropout outcomes were summarized as risk ratios when extractable. **Results:** The review included 65 reports from 2000–2026. Most reports were experimental (58/65, 89.2%), with crossover designs forming the largest detailed group (28/65, 43.1%). Sample size was extractable for 47 reports, totaling 1,799 participants. The most frequent breathing categories were hyperventilation (16/65, 24.6%), hypoventilation (14/65, 21.5%), paced breathing (13/65, 20.0%), and breath holding/apnea (10/65, 15.4%). Exploratory pooling showed small and uncertain effects for performance outcomes ($g = 0.09$, 95% CI -0.16 to 0.34) and physiological outcomes ($g = 0.21$, 95% CI -0.22 to 0.64). Safety and dropout risk-ratio syntheses did not show clear overall differences. **Conclusions:** Breathwork remains promising but methodologically uneven. The evidence supports more rigorous testing of sport-derived breathing routines, clearer intervention taxonomy, standardized outcome selection, and routine reporting of feasibility and adverse events.

Key takeaways

- The review includes 65 reports covering breathing techniques across sport, exercise, and performance contexts.
- Breathing interventions should not be treated as one uniform “breathwork” category.
- Pooled efficacy signals are small and uncertain, but the direction remains promising in several sport-relevant domains.
- Feasibility and safety reporting are still not strong enough for confident implementation guidance.
- Future studies should test empirically developed sport breathing routines with clear protocols, credible comparators, and complete reporting.

Keywords: scoping review; PRISMA-ScR; breathing training; breathwork; hyperventilation; hypoventilation; paced breathing; sport performance; feasibility; safety

1 Introduction

Breathing is automatic, but several of its parameters can be deliberately regulated. Respiratory frequency, depth, inhalation-exhalation ratio, breath-hold timing, route of breathing, and lung volume can all be modified intentionally. This makes breathing unusual among performance-relevant physiological functions: it is continuously available, requires little equipment, and can be applied before exercise, during exertion, during recovery, or as part of repeated training. In sport, athletes and practitioners use breath-regulation strategies to influence arousal, anxiety, oxygen saturation, carbon dioxide tolerance, acid-base balance, perceived exertion, attentional control, respiratory mechanics, and recovery.

The challenge is that “breathwork” is not one intervention. Hyperventilation, hypoventilation, paced breathing, breath holding, pranayama, diaphragmatic breathing, hook breathing, and breathing-pattern retraining have different mechanisms and risks. Some techniques aim to increase activation; others aim to down-regulate stress or support recovery. Some are single-session pre-performance routines; others are repeated training exposures. Some resemble sport-specific hypoxic or hypercapnic training, while others are better understood as psychophysiological regulation tools.

Previous synthesis has shown why this distinction matters. In a systematic review and meta-analysis of breathing techniques for physical sport performance, Laborde and colleagues separated slow-paced breathing, fast-paced breathing, voluntary hyperventilation, breath holding, and nostril-breathing approaches, and found that longer-term slow-paced breathing and breath-holding interventions showed more promising performance signals than short-term applications, while substantial heterogeneity and risk-of-bias concerns remained (Laborde et al., 2024). Their

review also highlighted a practical boundary that is important for sport: breathing techniques can often be performed in the field without devices that directly alter respiratory load or inspired gas composition, although biofeedback may be used to monitor or guide execution rather than to produce the respiratory stimulus itself.

This heterogeneity makes a scoping review appropriate. The purpose is not only to ask whether breathing techniques work, but to map what has been studied, how techniques have been categorized, which sport contexts and populations have been examined, which outcomes have been used, and where feasibility and safety reporting remain incomplete. This manuscript reports the evidence map, descriptive synthesis, exploratory effect syntheses, and safety and feasibility findings for the included literature.

2 Methods

2.1 Design and reporting

This manuscript is reported according to PRISMA-ScR principles. The review maps the available evidence rather than claiming a single definitive intervention effect. The unit of synthesis was the report or study row represented in the extraction table; quantitative syntheses used extractable effect rows from the accompanying meta-analysis dataset.

2.2 Eligibility concept

Eligible sources examined breathing techniques, breath-regulation routines, or breathing-related exposures relevant to sport, exercise, or performance. Eligible concepts included hyperventilation, hypoventilation, paced or slow breathing, breath holding/apnea, pranayama or yogic breathing, diaphragmatic breathing, hook breathing, Valsalva-type strategies, and related non-device approaches. Eligible contexts included athletes, physically active participants, laboratory exercise tasks, sport-specific testing, recovery, and performance-relevant psychophysiological outcomes.

2.3 Information sources and source selection

Searches were run in PubMed, Scopus, and SPORTDiscus, with additional reports identified through citation searching. After deduplication, title/abstract screening, full-text assessment, and citation backtracking, 65 reports were included. Figure 1 summarizes the study-selection flow.

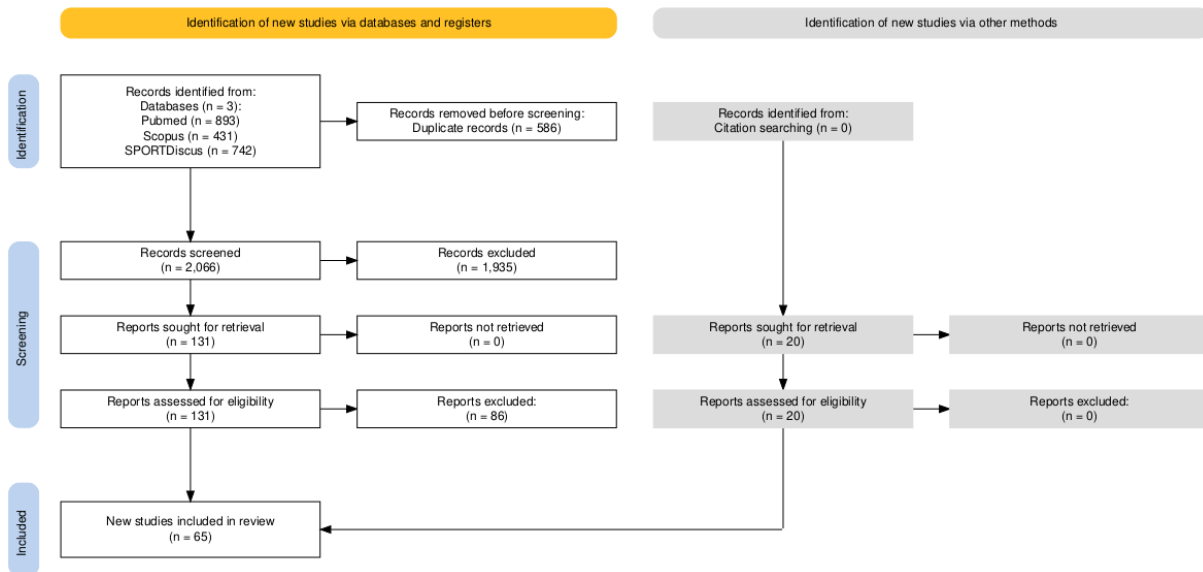


Figure 1: PRISMA flow diagram for study identification, screening, eligibility assessment, and inclusion.

Figure 1 shows that the database searches identified 2,066 records after duplicate removal. One hundred thirty-one reports were assessed for eligibility from database searching and 20 additional reports were assessed from other methods, leading to 65 included reports.

2.4 Data charting and quality control

Data were charted for bibliometry, study design, sport context, athlete level, sample size, age, sex distribution, baseline characteristics, breathing category, intervention dose, comparator, efficacy outcomes, uncertainty, feasibility, dropout, adverse events, and safety. Missing or unreported fields were retained as NR so that denominators and reporting gaps remained explicit.

2.5 Synthesis

Descriptive synthesis summarized publication year, study design, participant characteristics, sport context, breathing category, intervention duration, intervention frequency, feasibility, and safety. Exploratory meta-analysis used Hedges' g for efficacy outcomes when group-level data were extractable. Effect direction was standardized so that positive values favored the breathing intervention; lower-is-better outcomes such as time, reaction time, heart rate, RPE, dyspnea, fatigue, lactate, and decrement scores were reversed. Safety and dropout were synthesized as risk ratios where event data were available. Very imprecise safety estimates were retained in tables but excluded from displayed forest plots when the interval made the figure unreadable.

2.6 Critical appraisal

Formal risk-of-bias appraisal was not the primary objective of this scoping review. Instead, the synthesis emphasizes reporting completeness, design heterogeneity, missing feasibility fields, and uncertainty around quantitative estimates.

3 Results

The Results are organized to move from source selection and cohort description to the structure of the evidence map, then to intervention categories, efficacy signals, and feasibility and safety. This order reflects the purpose of a scoping review: first to define what the evidence base contains, and only then to interpret what the extractable outcomes suggest.

3.1 Study and participant characteristics

Table 1: Study-level baseline characteristics of the 65 included reports.

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Durand2000	2000	Obs. cross-sectional	Cycling, Triathlon, Running	elite	22	23.9 (1.3)	100
Spiering2004	2004	Obs. cross-sectional	Winter sports	elite	78	NR	32
Lindholm2005	2005	Obs. cross-sectional	Breath-hold diving	competitive	5	NR	NR
Woorons2008	2008	RCT	Running	competitive	16	27.1 (6.0)	NR
Guimard2014	2014	Acute experimental	Swimming	competitive	15	21.9 (0.9)	100
Malakhov2014	2014	Acute experimental	Field athletics, combative and team sports	competitive	66	19.8 (1.0)	44
Sakamoto2014	2014	Crossover	Cycling	NR	15	NR	NR
Fujii2015	2015	Crossover	Cycling	NR	NR	NR	NR
Jacob2015	2015	Crossover	Swimming	competitive	NR	NR	NR
Sakamoto2015	2015	Crossover	Isokinetic knee extension	competitive	15	21.2 (3.7)	73
Stavrou2015	2015	RCT	Fin-swimming	competitive	28	NR	50
Woorons2016	2016	RCT	Swimming (Triathlon)	competitive	16	NR	75
Burtch2017	2017	RCT	Swimming	competitive	20	NR	NR
Hakked2017	2017	Other experimental	Swimming	competitive	NR	NR	NR
Kacoglu2017	2017	Crossover	Jumping and sprinting	physically active	14	NR	NR
Pelka2017	2017	RCT	Sprinting	competitive	27	25.22 (1.08)	70
Trincat2017	2017	RCT	Swimming	competitive	16	15.55 (1.8)	56
Woorons2017	2017	Crossover	Cycling	competitive	9	27.2 (9.3)	89
FornasierSantos2018	2018	RCT	Rugby	competitive	35	NR	100
Gray2018	2018	Other experimental	Swimming	NR	NR	NR	NR
Lim2018	2018	Crossover	Cycling	physically active	10	24 (3)	100

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Table 1: Study-level baseline characteristics of the 65 included reports (continued).

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Sakamoto2018	2018	Crossover	Cycling	competitive	17	NR	88
Fernandez2019	2019	Obs. cross-sectional	Freediving	competitive	25	NR	100
Laborde2019	2019	Crossover	Burpees	physically active	120	25.57 (2.52)	58
Woorons2019	2019	Crossover	Shuttle run sprints	competitive	NR	NR	NR
Woorons2019b	2019	RCT	Cycling	competitive	18	34.6 (11)	100
Bouten2020	2020	Crossover	Cycling	physically active	11	NR	NR
Lapointe2020	2020	RCT	Basketball	competitive	17	22.3 (1.2)	71
Robertson2020	2020	Crossover	Swimming	competitive	NR	NR	NR
Sakamoto2020	2020	Crossover	Bench press, leg press	competitive	11	22.5 (4.3)	100
Woorons2020	2020	NR	NR	NR	NR	NR	NR
Bahensky2021	2021	RCT	NR	NR	NR	NR	NR
Citherlet2021	2021	Crossover	Repeated sprinting	competitive	NR	NR	NR
Dobashi2021	2021	Crossover	Cycling (Wingate)	competitive	11	24 (2)	91
Grossman2021	2021	Crossover	Cycling	physically active	10	23.7 (2.5)	100
Taylor2021	2021	Crossover	Memory-guided force pulse	healthy non-athlete	50	21.3 (2.8)	36
Vaidya2021	2021	NR	NR	NR	NR	NR	NR
Brocherie2022	2022	RCT	Ice hockey	elite	43	16.7 (1.5)	100
Conlon2022	2022	RCT	Shooting	competitive	64	NR	NR
Marko2022	2022	Before-after	Cycling	physically active	NR	NR	NR
You2023	2023	Crossover	NA	physically active	78	22.32 (2.11)	71
Braham2024	2024	RCT	Soccer	competitive	NR	NR	NR

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Table 1: Study-level baseline characteristics of the 65 included reports (continued).

	Study	Year	Design	Sport/context	Athlete level	N	Age	Male
							mean (SD)	(%)
∞	Buxton2024	2024	Crossover	Back Squat	recreationally trained	18	NR	100
	Li2024	2024	Crossover	Burpees	physically active	30	NR	NR
	Merlin2024	2024	RCT	Swimming	elite	13	13.9 (0.38)	38
	Rosa2024	2024	Crossover	Arm cycle ergometer	NR	14	NR	NR
	Sikora2024	2024	Obs. cross-sectional	NR	competitive	69	NR	NR
	Woorons2024	2024	RCT	Judo, Rowing	elite	20	20.6 (2.5)	50
	Deniz2025	2025	Crossover	Bench press	recreationally trained	12	27.92 (7.38)	100
	Fernandez2025	2025	Crossover	Freediving	competitive	13	42 (9)	NR
	Fessler2025	2025	Crossover	Sprint intervals	recreationally trained	36	NR	NR
	Fox2025	2025	RCT	General	physically active	404	37 (9.95)	44
	Kasap2025	2025	Crossover	HIIT	competitive	40	NR	NR
	Liapaki2025	2025	Crossover	Running	competitive	NR	24.7 (3.1)	NR
	Lorinczi2025	2025	RCT	Soccer	competitive	NR	NR	NR
	Yilmaz2025	2025	Other experimental	Wingate	physically active	NR	NR	NR
	Chiron2026	2026	RCT	Running	elite	22	22.8 (3.3)	68
	Devipriya2026	2026	Before-after	NA	healthy non-athlete	60	NR	50
	FernandezBarradas2026	2026	Before-after	American football	competitive	NR	NR	NR
	Iskra2026	2026	Crossover	NA	physically active	38	NR	NR

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Table 1: Study-level baseline characteristics of the 65 included reports (continued).

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Jing2026	2026	RCT	Swimming	competitive	NR	NR	NR
Jones2026	2026	RCT	Rowing	physically active	62	21.0 (3.0)	70
Katz2026	2026	Other experimental	Deadlift	physically active	40	NR	NR
Raidl2026	2026	RCT	Sprint interval training	competitive	NR	NR	NR
Tomita2026	2026	Crossover	HIIT	physically active	26	NR	NR

Note. NR = not reported; Obs. = observational. Values are shown as reported or extractable from the source papers.

Table 1 presents the included reports at study level before aggregation. This first table anchors the review in the actual studies rather than in summary statistics alone: the corpus spans more than two decades, multiple competitive and recreational contexts, and a mixture of observational, crossover, before-after, and randomized designs. The repeated NR entries are also informative, because they show where demographic or sample-size reporting was too incomplete to support clean denominators.

Table 2: Aggregate study and participant characteristics.

Characteristic	Summary
Reports in extraction corpus	65 reports
Publication years	2000–2026
Reports with extractable total sample size	47 reports
Aggregate reported sample size	1,799 participants
Age reporting	25 reports; weighted mean 28.1 years; pooled SD 10.1 years
Sex reporting	1,348 participants with sex data; estimated male proportion 59.8%
Athlete level	Competitive: 33; physically active: 14; elite: 6; recreationally trained: 3; healthy non-athlete: 2; not reported/blank: 7
Design profile	Experimental: 58 (89.2%); observational: 5 (7.7%); unclear: 2 (3.1%)

Table 2 summarizes the same corpus at aggregate level. The evidence base is dominated by experimental work, but this should not be read as uniform trial evidence: many reports use crossover, acute, or otherwise heterogeneous designs. The aggregate reported sample size was 1,799 participants, although sample size was not extractable in 18 rows, which limits precision when describing the population base.

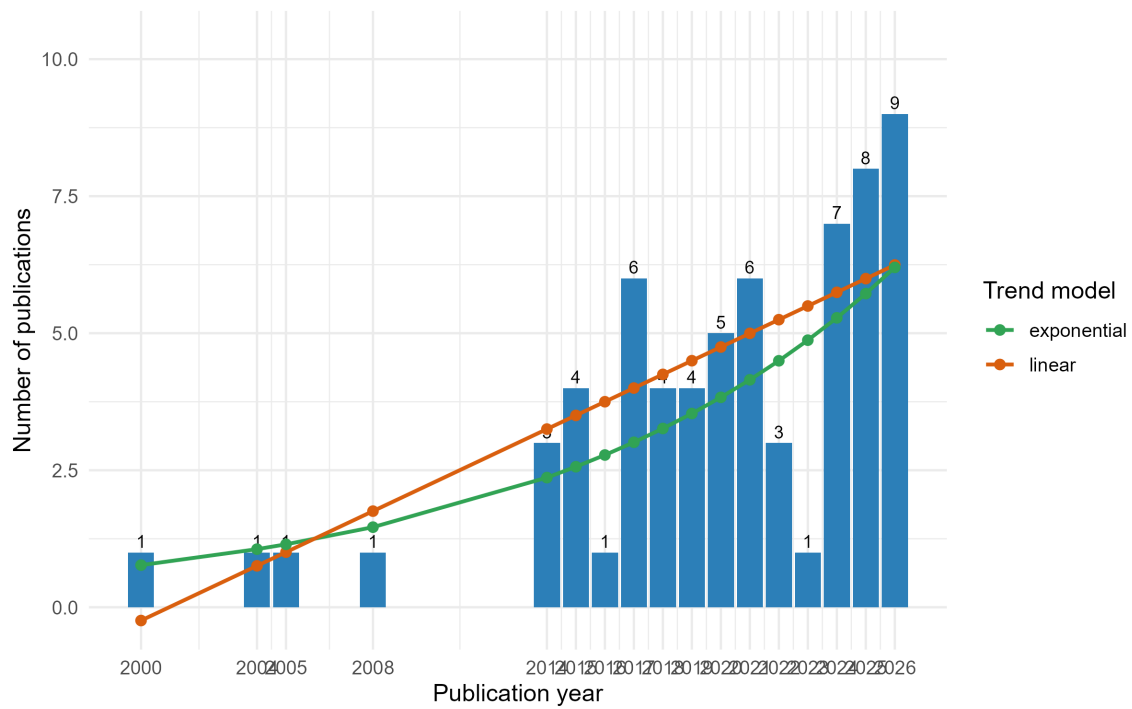


Figure 2: Annual publication counts and fitted trend lines.

Figure 2 shows a clear increase in publication volume after 2014, with the highest counts in 2024–2026. The linear trend estimated approximately 0.25 additional publications per year, while the exponential model had slightly better fit metrics. The field therefore appears to be expanding, but the publication curve reflects interest and availability of studies, not necessarily maturation of methods.

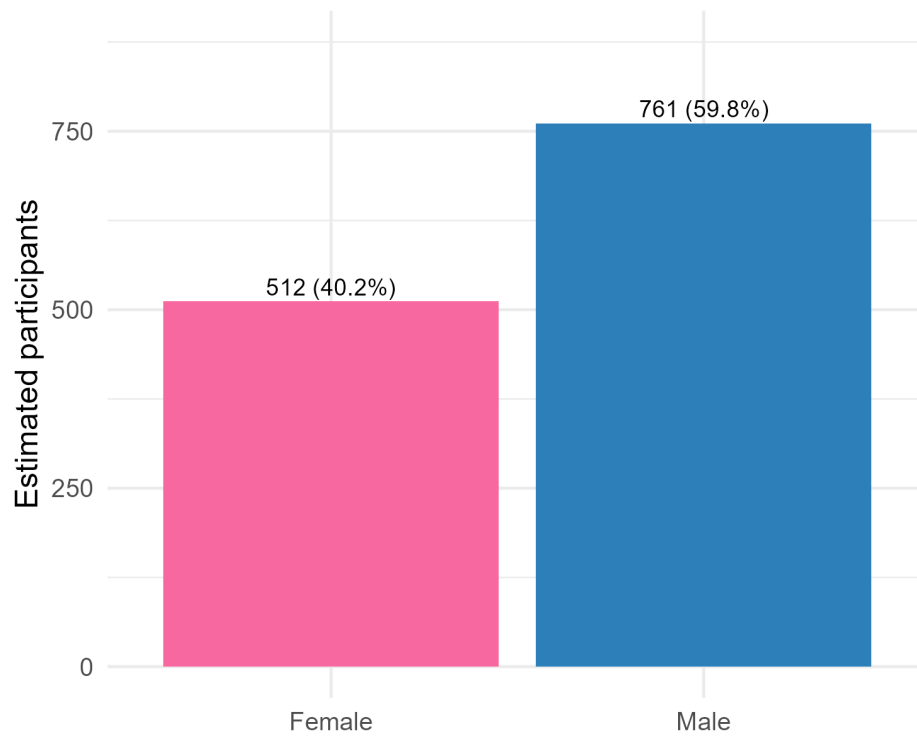


Figure 3: Estimated sex distribution among studies with extractable sex data.

Figure 3 indicates a male-predominant participant pool among reports with sex data. This imbalance matters for generalizability because breathing responses, sport participation patterns, and training contexts may differ across cohorts, yet several intervention categories remain underrepresented in female samples.

3.2 Evidence-map structure

Table 3: Evidence-map summary.

Domain	Main result
Detailed study designs	Crossover: 28 (43.1%); other/unclear experimental: 25 (38.5%); single-group/observational: 5 (7.7%); before-after: 3 (4.6%); acute experimental: 2 (3.1%); unclear: 2 (3.1%)
Most represented sport contexts	Cycling: 9 (13.8%); swimming: 9 (13.8%); unclear/not specified: 7 (10.8%); running: 3 (4.6%); burpees, freediving, HIIT, and soccer: 2 each (3.1%)
Breathing categories	Hyperventilation: 16 (24.6%); hypoventilation: 14 (21.5%); paced breathing: 13 (20.0%); breath holding/apnea: 10 (15.4%); other/unclear: 7 (10.8%); diaphragmatic: 2 (3.1%); pranayama/yogic: 2 (3.1%); hook breathing: 1 (1.5%)
Dose duration	33 parseable reports; median 1 day; IQR 1–28 days
Session frequency	23 parseable reports; median 1 session/week; IQR 1–2.5 sessions/week

Table 3 summarizes the key structural findings. The corpus is broad, but it remains uneven: many categories are represented by small clusters, and seven reports still required an other/unclear breathing label after broad classification. The map therefore supports a technique-specific interpretation rather than a single pooled claim about “breathwork”.

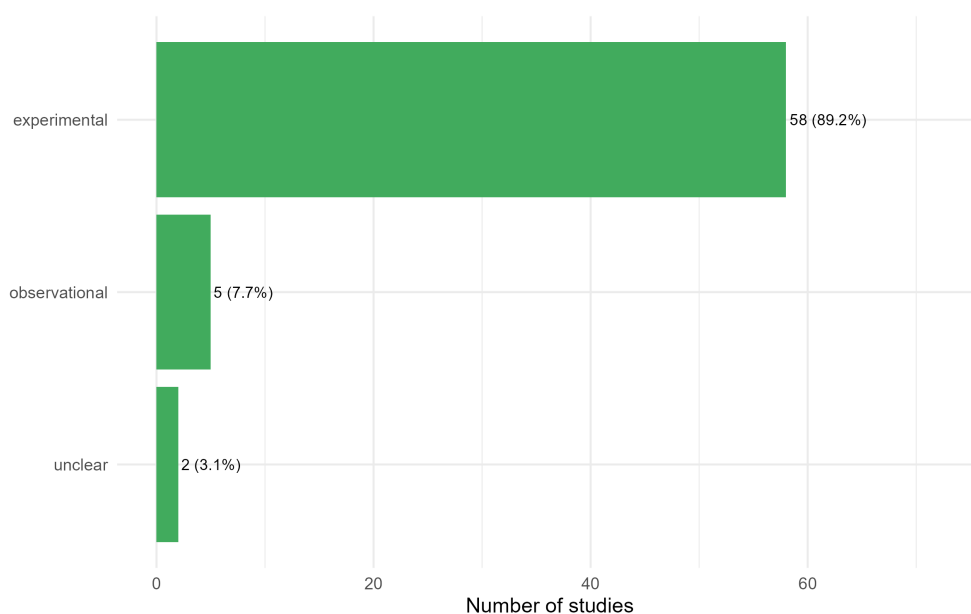


Figure 4: Broad study-design distribution.

Figure 4 shows that experimental reports dominate the corpus. This is encouraging for in-

tervention mapping, but it does not by itself imply high certainty because the detailed design distribution is more fragmented and often oriented toward acute laboratory effects.

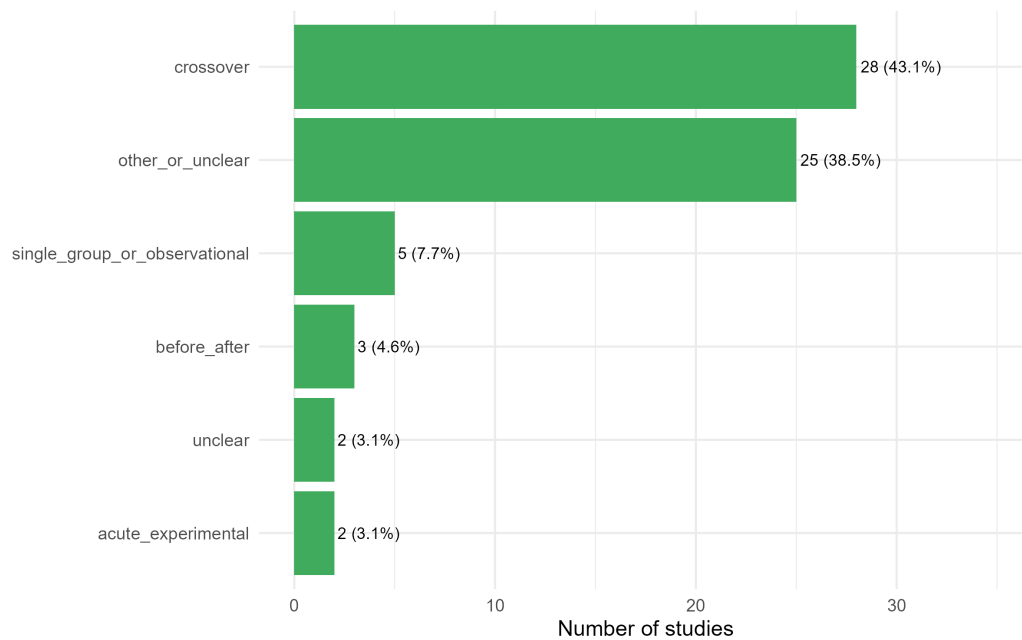


Figure 5: Detailed study-design distribution.

Figure 5 clarifies that crossover designs were the largest specific group, followed by other or unclear experimental designs. This pattern is consistent with a field that has many acute or laboratory-style studies but fewer large parallel-group trials able to test durable training effects or sport-specific implementation.

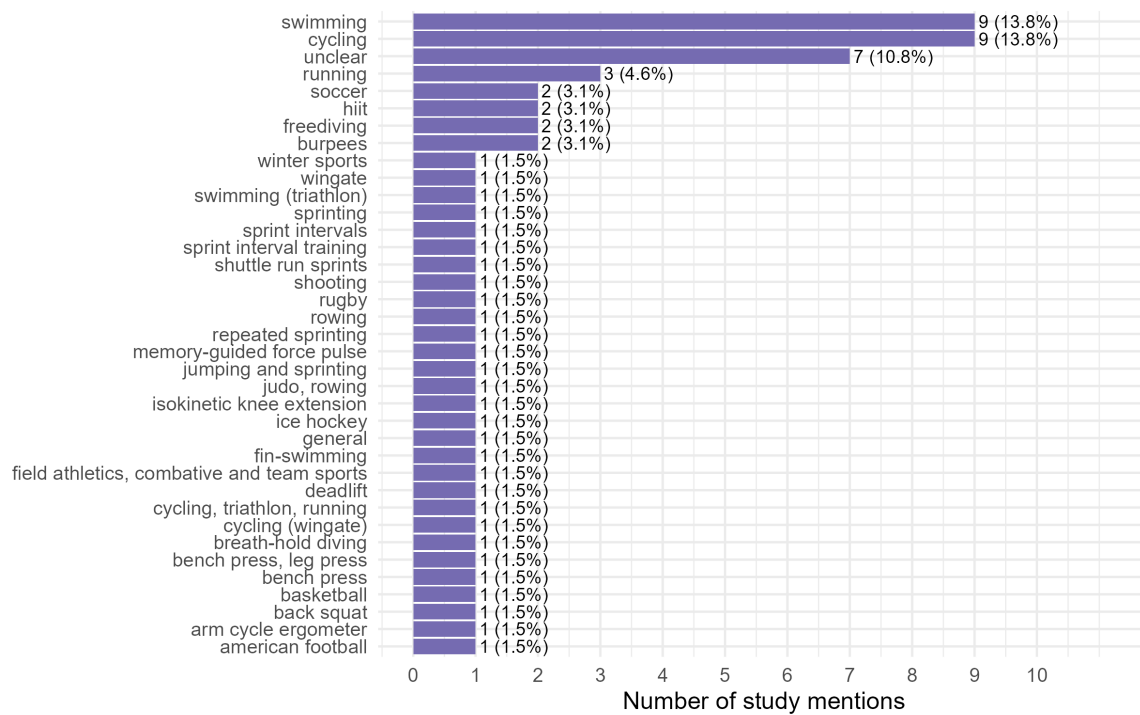


Figure 6: Distribution of sport and exercise contexts.

Figure 6 shows the sport-context dispersion. Cycling and swimming were the most represented contexts, but the long tail of single-study sports limits sport-specific certainty and makes it difficult to infer whether a technique tested in one task transfers to another sport.

3.3 Breathing categories and dose

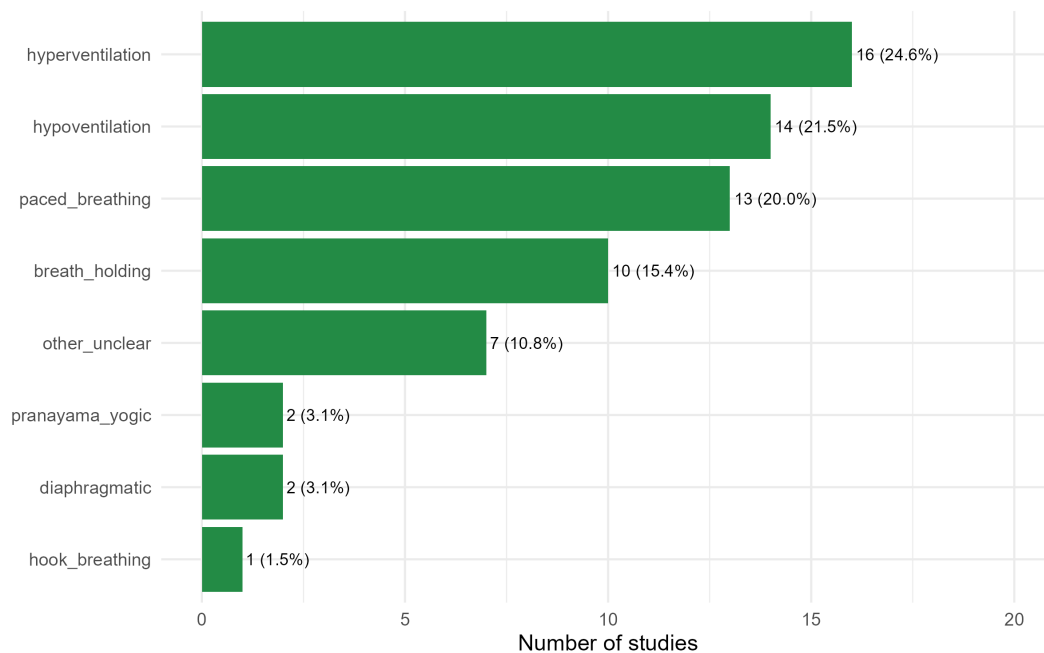


Figure 7: Breathing-category distribution.

Figure 7 reinforces the main conceptual point of the review: breathwork is a family of different interventions. Hyperventilation, hypoventilation, paced breathing, and breath holding together make up most of the corpus, but their mechanisms and safety profiles differ enough that they should be designed, tested, and interpreted separately.

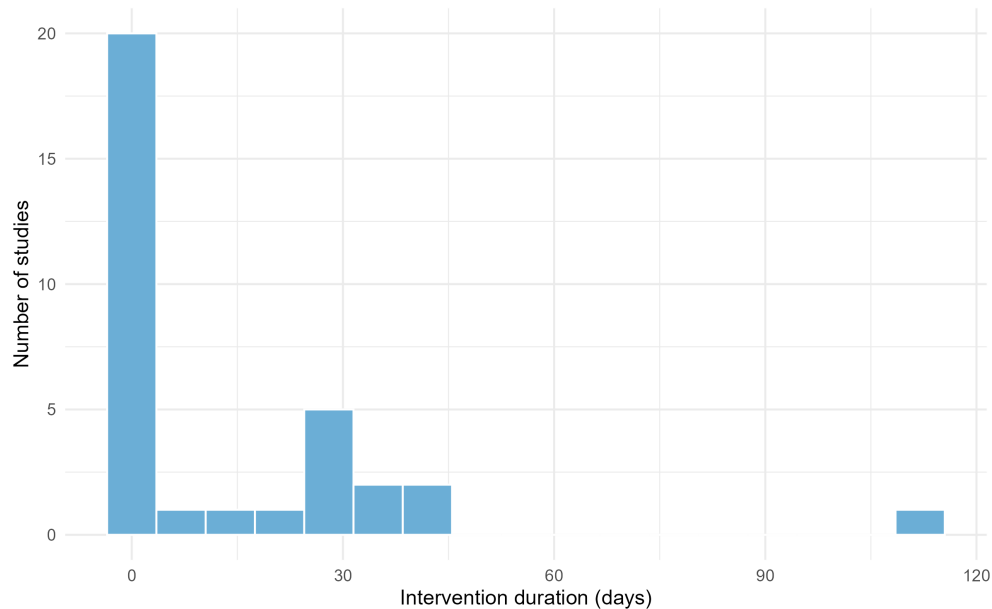


Figure 8: Intervention-duration distribution among reports with parseable duration.

Figure 8 shows that many interventions were acute or short-term. The median duration was one day, with a smaller subset extending across several weeks. This dose profile explains why the review can say more about immediate physiological or performance responses than about durable training adaptations.

3.4 Efficacy synthesis

Table 4: Exploratory quantitative syntheses.

Synthesis	k	Estimate	95% CI	I^2	Interpretation
Performance efficacy, Hedges' g	20	0.09	-0.16 to 0.34	70.9%	Small pooled direction favoring breathing intervention, but confidence interval crosses the null and prediction interval is wide.
Physiological efficacy, Hedges' g	17	0.21	-0.22 to 0.64	85.9%	Small-to-moderate pooled direction favoring breathing intervention, but uncertainty and heterogeneity are substantial.
Adverse events, risk ratio	29	1.00	0.96 to 1.04	0.0%	No clear overall difference in reported adverse-event risk; sparse-event methods and reporting limitations apply.
Dropout, risk ratio	33	0.91	0.59 to 1.41	0.0%	No clear difference in dropout risk between breathing and comparator conditions.

Table 4 reports the main exploratory quantitative syntheses. Performance and physiological outcomes both showed positive pooled directions, but neither confidence interval excluded the null. Heterogeneity was high, especially for physiological outcomes, so the pooled effects are best interpreted as weak signals that justify targeted trials and technique-specific reviews, not as definitive efficacy estimates.

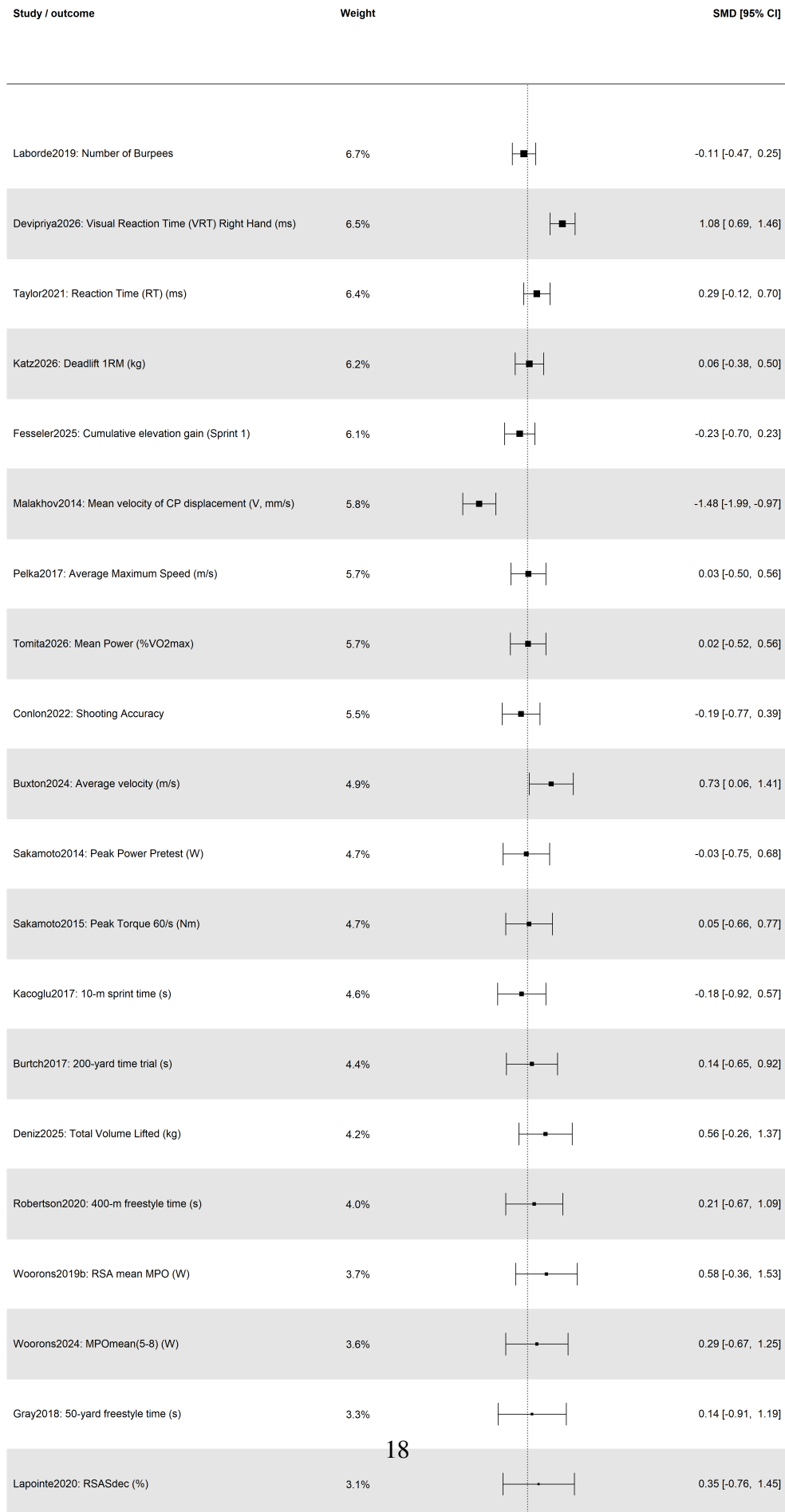
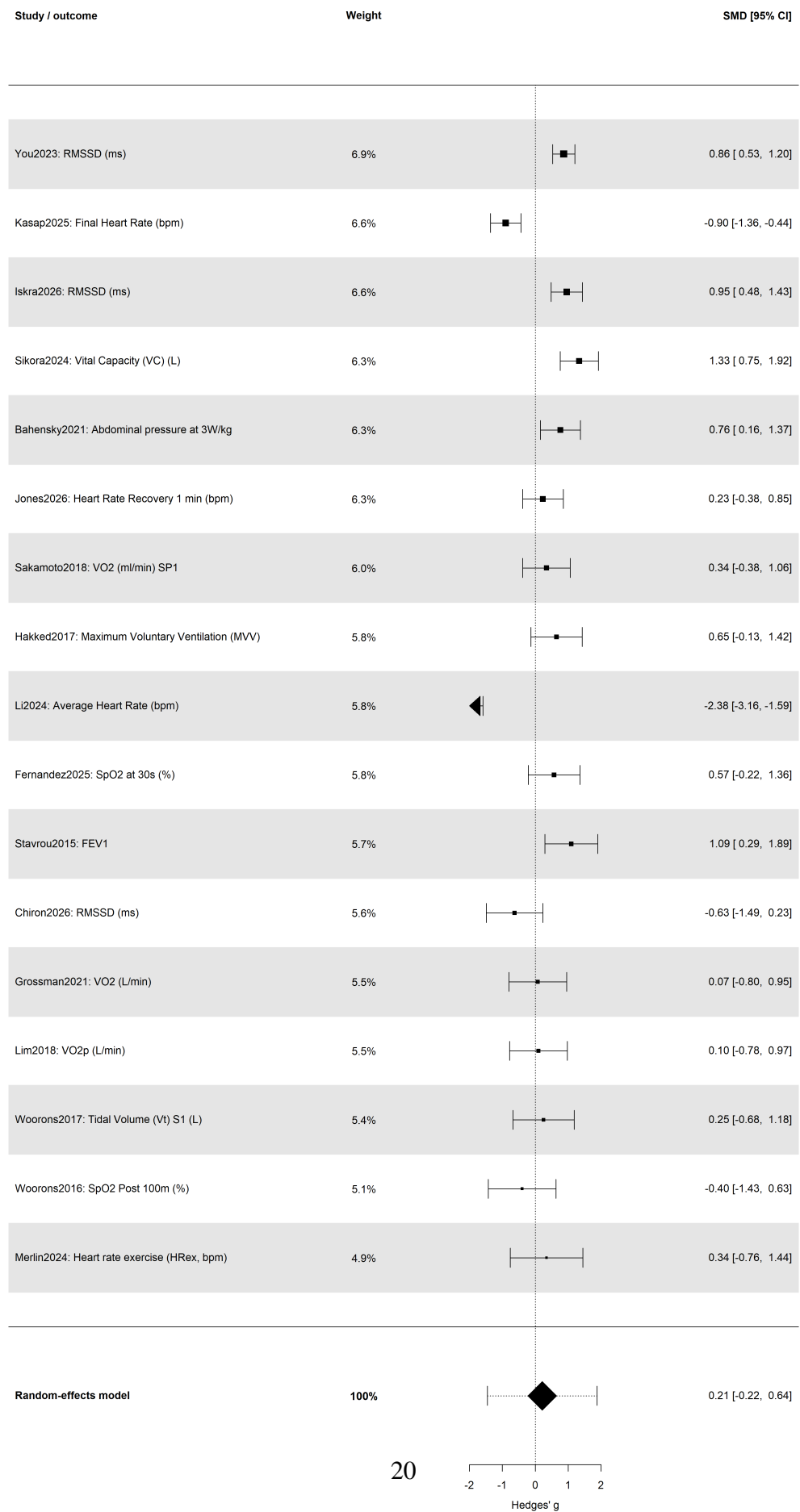


Figure 9 shows a mixed pattern across performance outcomes. Some reports favor the breathing intervention, while others are null or favor the comparator. The wide prediction interval is central to interpretation: a future study could plausibly show benefit, no effect, or harm depending on the technique, sport task, and outcome definition.



Heterogeneity: Q = 114.50, df = 16, p = 0.000; I² = 85.9%; tau² = 0.675

Figure 10 shows a slightly larger pooled direction than the performance forest plot, but heterogeneity is substantial. Physiological outcomes are mechanistically closer to breathing manipulation, yet they remain diverse: heart-rate, oxygenation, pulmonary-function, autonomic, and perceptual outcomes should not be treated as interchangeable.

3.5 Feasibility and safety

Table 5: Feasibility and safety reporting.

Field	Summary
Adherence	Only 2 reports had parseable adherence values; median adherence proportion 0.065. This should be interpreted as extraction/reporting sparsity rather than a representative adherence estimate.
Dropout	11 reports with parseable dropout; 871 participants with dropout data; pooled dropout proportion 20.9%.
Adverse-event mentions	Dizziness and tingling appeared in hyperventilation records; pulmonary edema appeared in breath-holding/hook-breathing contexts.
Displayed safety forest plot	Trincat2017 was excluded from the displayed safety risk-ratio plot because the upper confidence interval was extreme (RR upper CI 252.56), making the plot unreadable and uninformative. The study remained part of the extraction record.

Table 5 shows that feasibility and safety reporting remain major weaknesses. Dropout was more extractable than adherence, but adherence was rarely reported in a useful form. Safety reporting was sparse and likely underpowered for uncommon but relevant events, which is a serious limitation for interventions that are often promoted as simple and low risk.

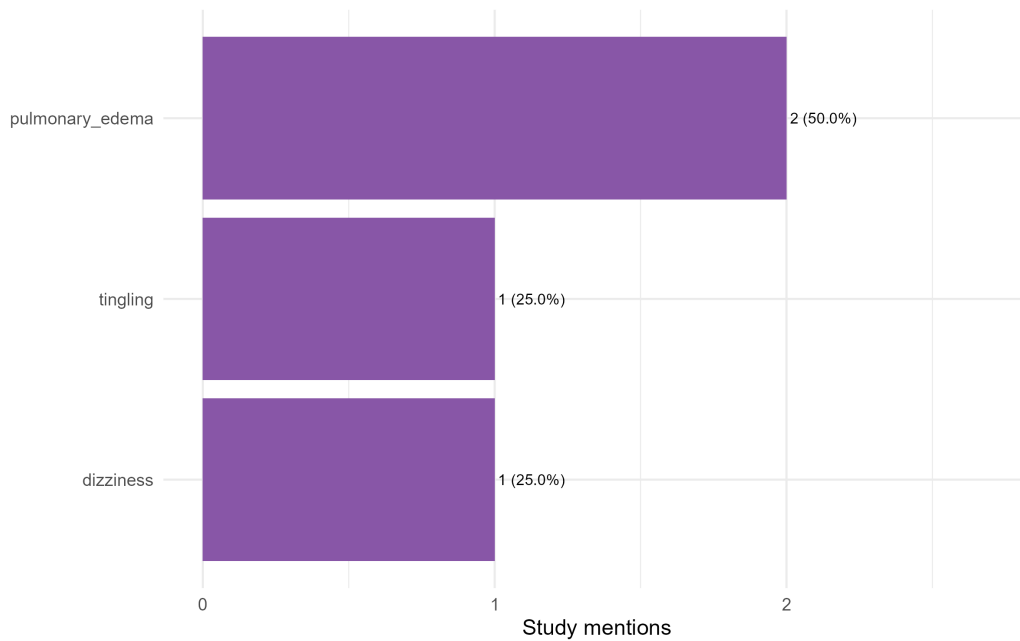


Figure 11: Adverse-event mentions by breathing-technique category.

Figure 11 shows that adverse-event mentions were few. Hyperventilation was associated with sensory symptoms such as dizziness and tingling, while pulmonary edema appeared in breath-holding or hook-breathing contexts. These findings are not incidence estimates, but they identify the technique families where safety monitoring and adverse-event definitions should be strengthened first.

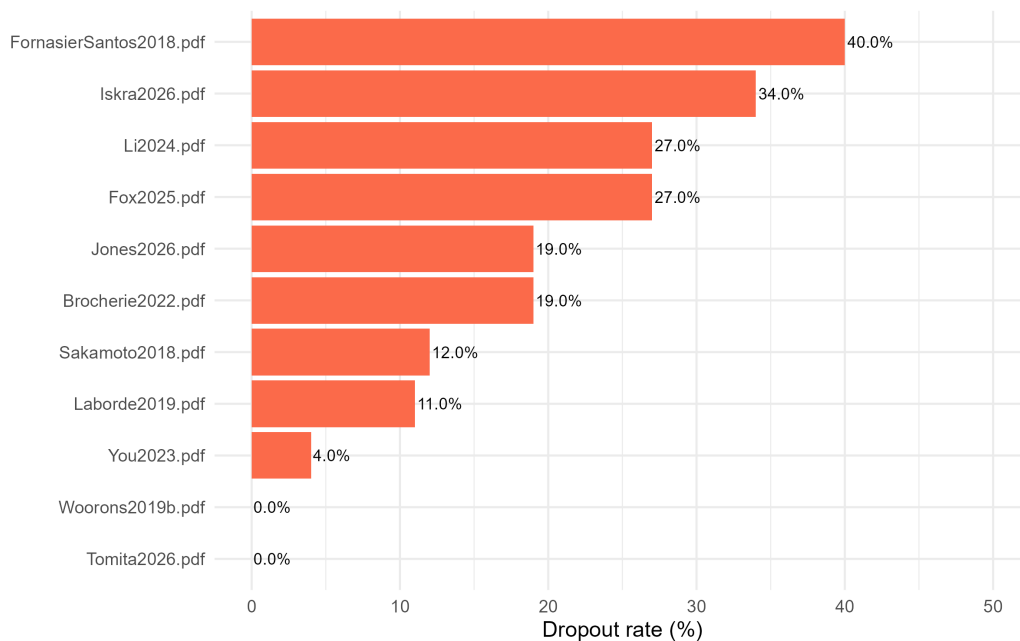


Figure 12: Dropout rates among reports with extractable dropout data.

Figure 12 shows that dropout reporting was available in only a subset of the corpus. The pooled dropout proportion was 20.9% among 871 participants with dropout data, but this should

not be generalized to all breathing interventions because missingness is substantial and reasons for discontinuation were not consistently described.

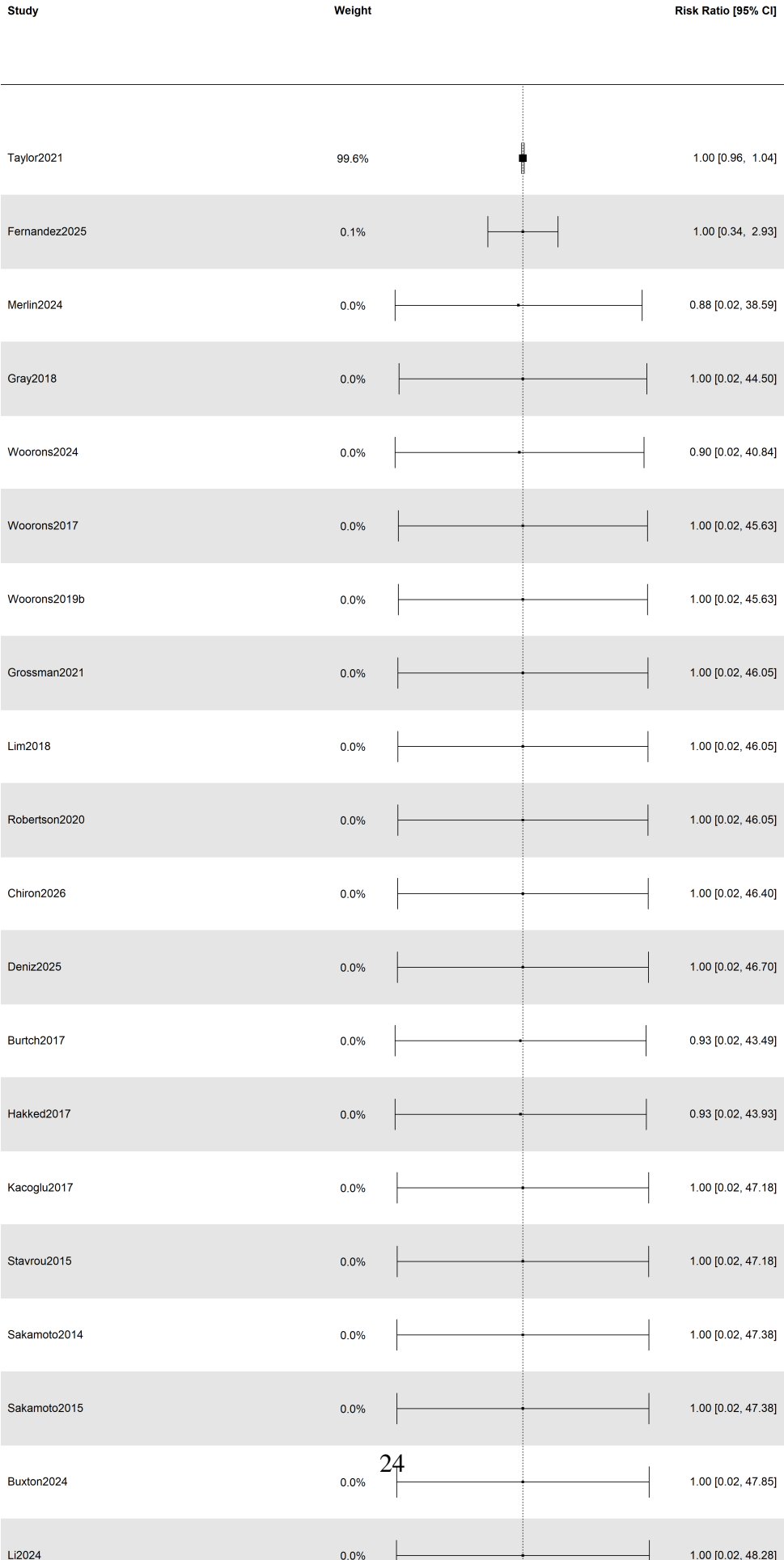


Figure 13 shows no clear pooled difference in adverse-event risk between breathing interventions and comparators. This should not be read as proof of safety; the event data are sparse, many studies were not designed primarily to detect adverse events, and rare risks may be invisible in small performance studies.

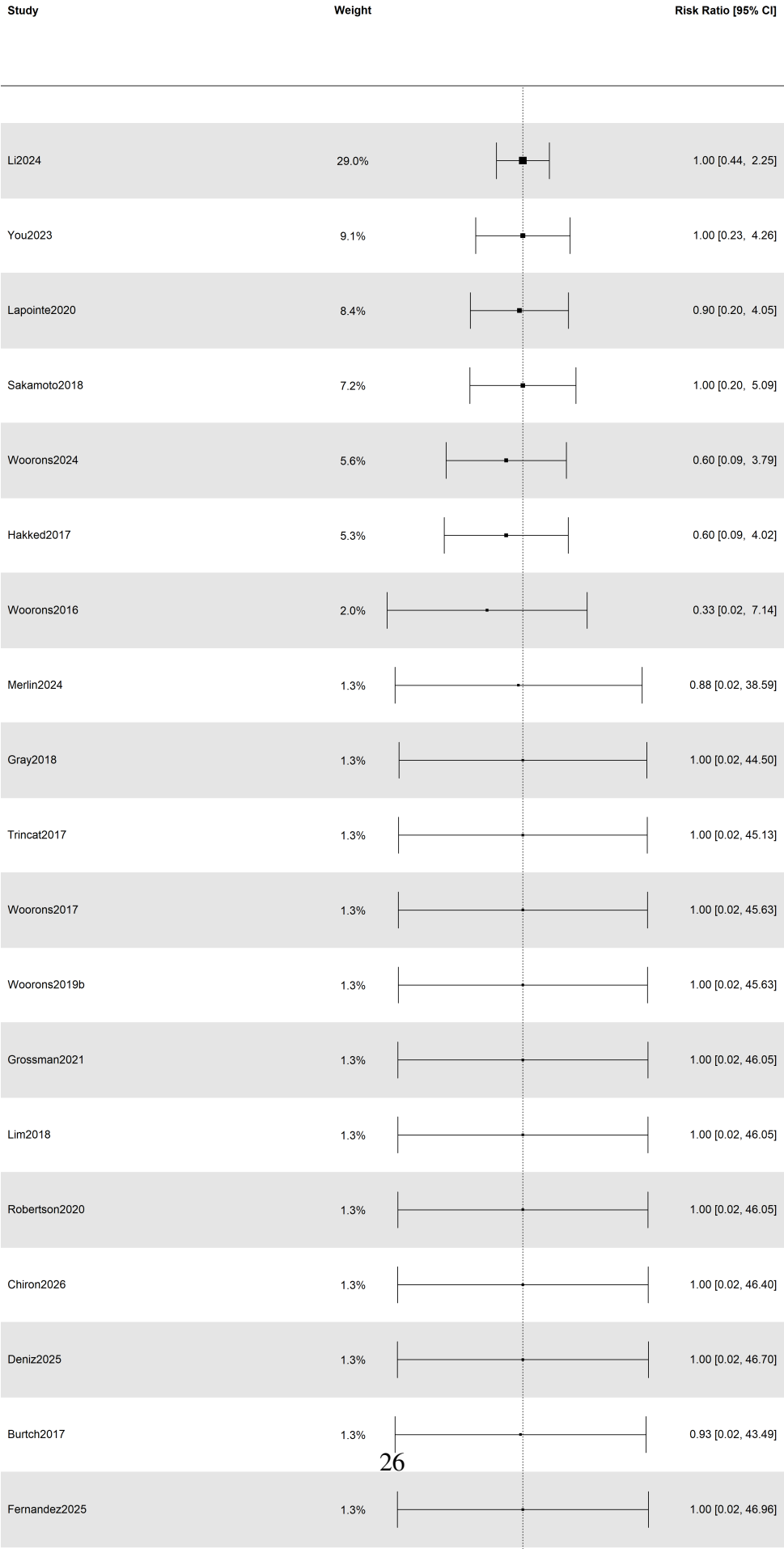


Figure 14 shows no clear overall difference in dropout risk. The estimate is compatible with either lower or higher dropout under breathing interventions, so future trials should report adherence, acceptability, and reasons for discontinuation as standard feasibility outcomes.

4 Discussion

4.1 Principal findings

This scoping review maps a heterogeneous but increasingly active field. The main conclusion is that breathwork in sport is promising but not mature. There are enough studies to identify recurring intervention families, plausible mechanisms, and outcome clusters, but not enough consistency to make broad claims that breathing interventions reliably improve performance across sports.

The evidence is strongest as a taxonomy and gap map. Hyperventilation, hypoventilation, paced breathing, breath holding, and pranayama/yogic breathing appear in different contexts and target different mechanisms. Pooling them as a single intervention would obscure more than it clarifies. The quantitative synthesis therefore separates performance, physiological, safety, and dropout outcomes, while retaining the broader scoping-review frame. The resulting narrative is consistent: interest is rising, protocols are diverse, efficacy signals are possible but uncertain, and feasibility and safety reporting have not kept pace with practical enthusiasm.

4.2 Contribution to the literature

The expanded extraction clarifies three points. First, the field is growing quickly, especially in the most recent years. Second, sport context remains dispersed, with cycling and swimming being the most represented but not dominant enough to anchor the whole field. Third, feasibility and safety reporting lag behind efficacy reporting. This is important because breathing routines are often promoted as low-cost and low-risk, but techniques that alter carbon dioxide, oxygen saturation, intrathoracic pressure, or autonomic state still require monitoring.

The review extends the existing systematic-review literature in a different direction. Laborde and colleagues asked an efficacy question and separated breathing techniques for meta-analysis, concluding that some longer-term protocols were promising but that high risk of bias, sparse technique-specific evidence, and heterogeneous protocols limited firm recommendations (Laborde et al., 2024). The present review starts from that same problem but maps a broader evidence base, including study-level characteristics, athlete level, sport context, intervention dose, feasibility, dropout, and safety reporting. This adds practical information that is not captured by efficacy estimates alone.

The review contributes by placing sport-related breathing techniques in a single structured map while preserving clinically and practically important distinctions between intervention fam-

ilies. It also standardizes effect direction before exploratory pooling. Direction standardization is essential: for outcomes such as final heart rate, reaction time, sprint time, RPE, or decrement score, lower values may represent better performance or recovery. Without this step, pooled Hedges' g can be misleading.

4.3 Implications for practice

Practitioners should use breathing techniques as targeted tools rather than generic breathwork. Slow or paced breathing may be most relevant for regulation, recovery, or attentional control. Hyperventilation may be relevant in short high-intensity contexts but requires attention to symptoms and safety. Hypoventilation and breath-holding interventions belong closer to hypoxic or repeated-sprint training logic and need careful supervision. Hook breathing and freediving-related strategies have distinct risk considerations.

4.4 Implications for research

New sport-derived breathing routines should be tested rigorously instead of being adopted because they are plausible or popular. Future trials should prespecify the breathing rate, depth, breath-hold timing, lung volume, total exposure, comparator, and expected mechanism. They should also report adherence, dropout, acceptability, adverse events, and reasons for discontinuation. For efficacy, researchers should select outcomes that match the proposed mechanism and report group-level means and dispersions so that future meta-analyses can be less dependent on narrative extraction.

4.5 Limitations

Several limitations should be considered. Some reports did not provide complete sample-size, demographic, intervention-dose, feasibility, or adverse-event data, and these gaps limit denominator certainty. Quantitative syntheses are exploratory and combine outcome domains that remain heterogeneous. The safety and dropout analyses are limited by sparse events and incomplete reporting. The review should therefore be interpreted primarily as an evidence map and hypothesis-generating synthesis rather than as a definitive estimate of intervention efficacy.

5 Conclusions

Breathing techniques in sport and performance contexts are a growing and promising field, but the evidence is still fragmented. Exploratory performance and physiological pooled effects were positive but uncertain. Safety and dropout syntheses did not show clear overall excess risk, but reporting was too incomplete to support strong reassurance. The next phase of research should

move from informal sport-derived routines toward clear protocols, credible controls, complete feasibility reporting, and mechanism-matched outcomes.

Declarations

Funding: No external funding was declared.

Conflicts of interest: The authors declare no conflicts of interest.

Data availability: The extraction data, generated tables, plots, and scripts are available in the project repository.

Author contributions: The authors contributed to study conception, data charting, analysis, interpretation, and manuscript preparation.

A PRISMA-ScR Checklist Crosswalk

Table 6: PRISMA-ScR checklist crosswalk.

Item	Reporting location	Status
Title and abstract	Title and structured abstract identify the manuscript as a scoping review and summarize source selection, charting, and synthesis.	Reported
Rationale	Introduction explains why heterogeneous breathing strategies require mapping before definitive efficacy recommendations.	Reported
Objectives	Introduction states mapping objectives for study characteristics, interventions, outcomes, feasibility, safety, and exploratory effects.	Reported
Eligibility criteria	Methods describe eligible breathing techniques, populations, and sport/performance contexts.	Reported
Information sources	Methods identify the databases, citation searching, and extraction dataset.	Reported
Selection of sources	Results include Figure 1.	Reported
Data charting	Methods describe bibliometric, participant, intervention, outcome, feasibility, and safety fields.	Reported
Synthesis of results	Results combine descriptive mapping, plots, and exploratory effect synthesis.	Reported
Critical appraisal	Methods state that formal risk-of-bias appraisal was not the central scoping-review objective; reporting completeness is summarized instead.	Reported

Item	Reporting location	Status
Limitations	Discussion describes database/source-set, extraction, reporting, heterogeneity, and meta-analysis limitations.	Reported
Funding and conflicts	Declarations section included.	Reported

Table 6 provides a reporting crosswalk for the main PRISMA-ScR items. It is included to make the review structure transparent and to show where selection, charting, synthesis, limitations, and declarations are addressed in the manuscript.

B Exact Search Strings

Pubmed

```
(
  (
    "Healthy Volunteers" [Mesh]
    OR "Athletes" [Mesh]
    OR healthy [Title/Abstract]
    OR "healthy volunteer*" [Title/Abstract]
    OR athlete* [Title/Abstract]
    OR sport* [Title/Abstract]
    OR "physically active" [Title/Abstract]
    OR "recreationally active" [Title/Abstract]
  )
  AND
  (
    "breath-holding" [Title/Abstract]
    OR "breath holding" [Title/Abstract]
    OR "breath hold*" [Title/Abstract]
    OR "breath-hold*" [Title/Abstract]
    OR apnea [Title/Abstract]
    OR apnoea [Title/Abstract]
    OR "breathing exercise*" [Title/Abstract]
    OR "controlled breathing" [Title/Abstract]
    OR "controlled frequency breathing" [Title/Abstract]
    OR "paced breathing" [Title/Abstract]
    OR "paced respiration" [Title/Abstract]
    OR "metronome breathing" [Title/Abstract]
  )
)
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OR "metronome-paced breathing"[Title/Abstract]
OR "coherent breathing"[Title/Abstract]
OR "coherence breathing"[Title/Abstract]
OR "resonance breathing"[Title/Abstract]
OR "resonance frequency breathing"[Title/Abstract]
OR "slow breathing"[Title/Abstract]
OR "slow-paced breathing"[Title/Abstract]
OR "deep breathing"[Title/Abstract]
OR "diaphragmatic breathing"[Title/Abstract]
OR "abdominal breathing"[Title/Abstract]
OR "fast breathing"[Title/Abstract]
OR "fast-paced breathing"[Title/Abstract]
OR "rapid breathing"[Title/Abstract]
OR hyperventilat*[Title/Abstract]
OR hypocapni*[Title/Abstract]
OR hypoventilat*[Title/Abstract]
OR pranayama[Title/Abstract]
OR prana*[Title/Abstract]
OR "nostril breathing"[Title/Abstract]
OR "alternate nostril"[Title/Abstract]
OR "alternate-nostril"[Title/Abstract]
OR "uni-nostril"[Title/Abstract]
OR "unilateral nostril"[Title/Abstract]
OR "Wim Hof"[Title/Abstract]
OR "HRV biofeedback"[Title/Abstract]
OR "heart rate variability biofeedback"[Title/Abstract]
OR (
  biofeedback[Title/Abstract]
  AND (
    breath*[Title/Abstract]
    OR respiration[Title/Abstract]
    OR respiratory[Title/Abstract]
    OR "heart rate variability"[Title/Abstract]
    OR HRV[Title/Abstract]
    OR "resonance frequency"[Title/Abstract]
  )
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AND

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(
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  OR competition[Title/Abstract]
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  OR power[Title/Abstract]
  OR strength[Title/Abstract]
  OR "time trial"[Title/Abstract]
  OR "repeated sprint"[Title/Abstract]
)
)
NOT
(
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  OR "Respiratory Muscle Training"[Title/Abstract]
  OR "Inspiratory Muscle Training"[Title/Abstract]
  OR COPD[Title/Abstract]
  OR asthma[Title/Abstract]
  OR "sleep apnea"[Title/Abstract]
  OR "sleep apnoea"[Title/Abstract]
  OR stroke[Title/Abstract]
  OR "pulmonary disease"[Title/Abstract]
  OR dyspnea[Title/Abstract]
  OR dyspnoea[Title/Abstract]
)
)
```

Scopus

```
TITLE-ABS-KEY (
  athlete* OR sport* OR "physically active" OR "recreationally active"
)
)
```

AND

```
TITLE-ABS-KEY (
  "breath holding" OR "breath-holding" OR "breath hold*" OR "breath-hold*" OR
  apnea OR apnoea OR
  "breathing exercise*" OR "controlled breathing" OR "controlled frequency breathing"
  "paced breathing" OR "paced respiration" OR
  "slow breathing" OR "slow-paced breathing" OR "coherent breathing" OR "coherence br
  "resonance breathing" OR "resonance frequency breathing" OR
)
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"diaphragmatic breathing" OR "abdominal breathing" OR "deep breathing" OR
"fast breathing" OR "fast-paced breathing" OR "rapid breathing" OR
hyperventilat* OR hypocapni* OR hypoventilat* OR
pranayama OR "yogic breathing" OR
"nostril breathing" OR "alternate nostril" OR "alternate-nostril" OR
"Wim Hof" OR "box breathing" OR "hook breathing" OR
"HRV biofeedback" OR "heart rate variability biofeedback"
)
AND
TITLE-ABS-KEY (
performance OR endurance OR competition OR sprint* OR power OR strength OR
"time trial" OR "repeated sprint*" OR "athletic performance" OR
"sport performance" OR "sports performance"
)
AND NOT
TITLE-ABS-KEY (
"respiratory muscle training" OR "inspiratory muscle training" OR
COPD OR asthma OR "sleep apnea" OR "sleep apnoea" OR stroke OR
"pulmonary disease" OR dyspnea OR dyspnoea
)
AND NOT DOCTYPE(re)

SPORTDiscus
(
TI (
athlete* OR sport* OR "physically active" OR "recreationally active" OR trained OR
)
OR AB (
athlete* OR sport* OR "physically active" OR "recreationally active" OR trained OR
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)
AND
(
TI (
"breath holding" OR "breath-holding" OR "breath hold*" OR "breath-hold*" OR apnea
OR "breathing exercise*" OR "controlled breathing" OR "controlled frequency breathi
OR "paced breathing" OR "paced respiration" OR "metronome breathing" OR "metronom
OR "slow breathing" OR "slow-paced breathing" OR "coherent breathing" OR "coheren
OR "resonance breathing" OR "resonance frequency breathing"

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OR "diaphragmatic breathing" OR "abdominal breathing" OR "deep breathing"
OR "fast breathing" OR "fast-paced breathing" OR "rapid breathing"
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OR pranayama OR prana* OR "yogic breathing"
OR "nostril breathing" OR "alternate nostril" OR "alternate-nostril" OR "uni-nost.
OR "Wim Hof" OR "box breathing" OR "hook breathing"
OR "HRV biofeedback" OR "heart rate variability biofeedback"
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OR AB (
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OR "breathing exercise*" OR "controlled breathing" OR "controlled frequency breat
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OR "slow breathing" OR "slow-paced breathing" OR "coherent breathing" OR "coheren
OR "resonance breathing" OR "resonance frequency breathing"
OR "diaphragmatic breathing" OR "abdominal breathing" OR "deep breathing"
OR "fast breathing" OR "fast-paced breathing" OR "rapid breathing"
OR hyperventilat* OR hypocapni* OR hypoventilat*
OR pranayama OR prana* OR "yogic breathing"
OR "nostril breathing" OR "alternate nostril" OR "alternate-nostril" OR "uni-nost.
OR "Wim Hof" OR "box breathing" OR "hook breathing"
OR "HRV biofeedback" OR "heart rate variability biofeedback"
)
)
AND
(
    TI (
        performance OR endurance OR competition OR sprint* OR power OR strength OR "time
        OR "repeated sprint*" OR recovery OR fatigue OR "athletic performance" OR "sport
    )
    OR AB (
        performance OR endurance OR competition OR sprint* OR power OR strength OR "time
        OR "repeated sprint*" OR recovery OR fatigue OR "athletic performance" OR "sport
    )
)
NOT
(
    TI (
        review OR "systematic review" OR "scoping review" OR meta-analysis OR "meta analy
        OR "respiratory muscle training" OR "inspiratory muscle training"
    )
)

```

OR COPD OR asthma OR "sleep apnea" OR "sleep apnoea" OR stroke OR "pulmonary dise
)
 OR AB (
 "systematic review" OR "scoping review" OR meta-analysis OR "meta analysis"
 OR "respiratory muscle training" OR "inspiratory muscle training"
 OR COPD OR asthma OR "sleep apnea" OR "sleep apnoea" OR stroke OR "pulmonary dise
)
)

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